

scripts/autonomous-qa/lib-emulator.sh — User Guide

Last verified: 2026-07-26 (LVA-018 script-docs backfill cycle)

Inheritance: HelixConstitution §11.4.18 (script docs); Lava §6.AG / §6.AH / §6.X (containerized emulators), §6.R (no-hardcoding, slirp exemption) **Classification:** project-specific

Overview

Sourceable library of thin Lava glue replicating submodules/containers/pkg/emulator/containerized.go to boot the §6.X **containerized KVM Android emulator** image and wire host adb. The emulator process runs INSIDE a podman container (image built from the submodule Containerfile), NEVER host-direct, NEVER a live device. The image bundles the AVD default, an adb keypair, and a socat bridge (container 0.0.0.0:5575 → 127.0.0.1:5555 adbd).

Functions

Function	Purpose
emu_pick_port	Pick a free TCP port on loopback
emu_boot [avd]	Boot the container (cold boot); prints the container name; persists state
emu_authorize_adb <container>	Copy the image's baked adb key out and point the host adb client at it (ADB_VENDOR_KEYS)
emu_connect <adb_port>	adb connect; prints the adb serial (127.0.0.1:<port>)
emu_wait_boot <serial> [timeout_secs]	Poll sys.boot_completed + pm ready (default 360 s); dumps container logs on timeout
emu_fix_network <serial>	Ensure the guest has an active default network (see below)
emu_cleanup_orphans	Remove leftover lava-emu-* containers from an interrupted run
emu_tearardown [container]	Remove the container + the state file

State + configuration

- State file: scripts/autonomous-qa/.emu-state (CONTAINER / ADB_PORT / CONSOLE_PORT) — written by emu_boot, removed by emu_teardown.
- EMU_IMAGE (env, default ghcr.io/vasic-digital/lava-android-emulator:api34-x86_64) — the emulator image.
- Container run flags: --userns=keep-id --device /dev/kvm, host-loopback port publishing only, ANDROID_COLD_BOOT=true.

emu_fix_network — the eth0/ndc default-network fix

On some hosts (observed: podman 4.9.3 on thinker.local) the emulator boots with eth0 DOWN and Android never registers it as a framework network — apps see "Network unreachable" because Android routes per-network via netd mark-based tables. The function (root via su 0 inside the guest) brings eth0 up and adds it to the default physical network (netId 100) through ndc, retrying up to 8 times until the slirp gateway pings. Self-gating + idempotent: if the gateway already answers it no-ops; it never fails the run — the iteration records the REAL outcome (anti-bluff: a stalled onboarding shows as SKIP/FAIL).

§6.R note: the 10.0.2.x literals are platform-FIXED Android-emulator QEMU user-mode (slirp) constants hardwired by the emulator — 10.0.2.2 = host-loopback alias, 10.0.2.3 = built-in DNS, 10.0.2.15 = the guest NIC. They are the emulator equivalent of 127.0.0.1 and cannot drift; the §6.R IPv4 scanner carries a narrow path+range exemption for the 10.0.2.0/24 slirp range scoped to this helper (see scripts/scan-no-hardcoded-ipv4.sh).

Usage

```
source scripts/autonomous-qa/lib-emulator.sh
emu_cleanup_orphans
CONTAINER="$(emu_boot default)"
ADB_PORT="$(grep '^ADB_PORT=' scripts/autonomous-qa/.emu-state |
    cut -d= -f2)"
emu_authorize_adb "$CONTAINER"
SERIAL="$(emu_connect "$ADB_PORT")"
emu_wait_boot "$SERIAL" 360
emu_fix_network "$SERIAL"
# ... run tests against $SERIAL ...
emu_teardown
```

Companion files

- `scripts/autonomous-qa/run-matrix.sh`, `run-nav-challenges.sh`, `release-coldstart-canary.sh` — the consumers
- `submodules/containers/pkg/emulator/` — the canonical Go implementation this mirrors
- `docs/scripts/run-emulator-tests.sh.md` — the sibling Containers-driven emulator runner